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EventHub: Data Factory for Generalizable Event-Based Stereo Networks without Active Sensors

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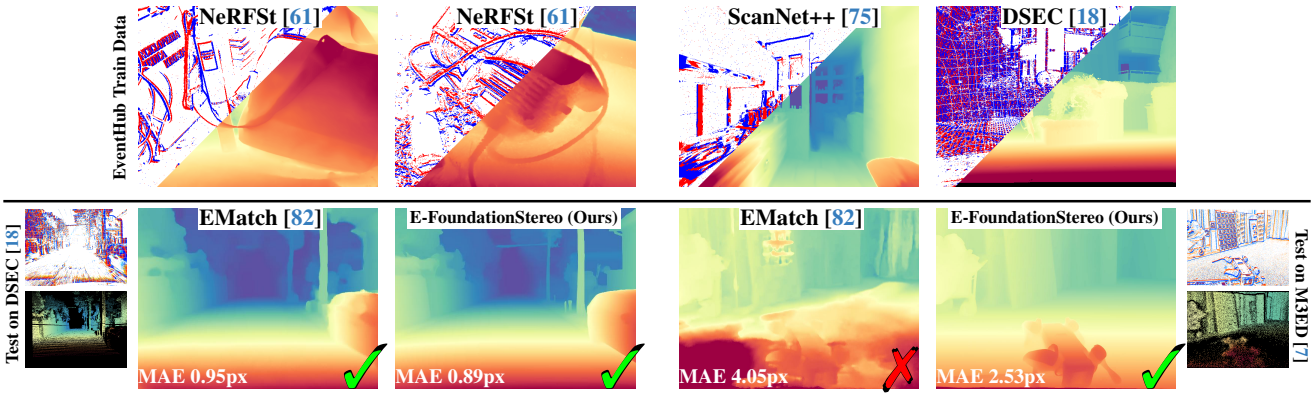


Figure 1. **EventHub: LiDAR-free proxy data for robust event stereo.** Our factory generates training data from multiple sources [18, 61, 75] (top), allowing our E-FoundationStereo to match EMatch [82] in-domain [18] and outperform it in generalization [7] (bottom).

Abstract

We propose EventHub, a novel framework for training deep-event stereo networks without ground truth annotations from costly active sensors, relying instead on standard color images. From these images, we derive either proxy annotations and proxy events through state-of-the-art novel view synthesis techniques, or simply proxy annotations when images are already paired with event data. Using the training set generated by our data factory, we repurpose state-of-the-art stereo models from RGB literature to process event data, obtaining new event stereo models with unprecedented generalization capabilities. Experiments on widely used event stereo datasets support the effectiveness of EventHub and show how the same data distillation mechanism can improve the accuracy of RGB stereo foundation models in challenging conditions such as nighttime scenes.

1. Introduction

Now nearing its fiftieth anniversary [63], stereo matching has undergone rapid evolution over the past decade thanks to deep learning [63], thus enabling high-accuracy and high-resolution depth maps that are crucial for applications such as autonomous driving, 3D scene reconstruction, augmented reality, and robotic navigation. Recent deep-based stereo models achieved remarkable performance [6, 69], also in a zero-shot manner, thanks to a large quantity of labeled data – i.e., millions of labeled synthetic and real images. The acquisition of those images required years of incredible efforts from the community: starting from sophisticated active setups [56] to achieve high-accuracy real datasets, to the usage of large computing resources to render large-scale photo-realistic synthetic datasets [69].

Recently, the introduction of the first commercial event cameras [39] led to the creation of a novel branch of stereo literature that aims to estimate depth from a pair of synchronized event cameras [20]. These sensors capture asynchronous per-pixel brightness changes occurring in the scene, so-called "events" [15, 53]. An event is characterized by a pixel coordinate (the location where the change occurred), a timestamp (when it occurred), and a polarity (± 1, indicating whether brightness increased or decreased). The asynchronous working principle enables these sensors to capture information at microsecond resolution, allowing them to surpass traditional frame-based cameras in chal-

lenging scenarios, such as fast motion (resulting in no motion blur) and high dynamic (resulting in no over/under-exposure). As a drawback, adapting the large body of image-based computer vision algorithms to event cameras is not trivial due to the very sparse nature of events [15].

Despite the growing interest in event-based stereo matching, the availability of labeled datasets remains very limited compared to the traditional frame-based domain [20]. Capturing dense and accurate ground truth for asynchronous event streams is significantly more challenging due to the still-emergent event community and the substantial deviation from traditional frame-based cameras.

In this paper, we aim to introduce a novel framework for training deep-based event stereo networks effortlessly and without any ground-truth. By leveraging state-of-the-art novel view synthesis solutions [59], we can generate event stereo training data from image sequences collected with a single color camera, alongside with proxy depth labels. In alternative, when paired RGB stereo images and event stereo data are available, we distill the knowledge of stereo foundation models processing to annotate the latter. This approach drastically reduces the need for complex data acquisition setups and large-scale manual labeling efforts, democratizing access to high-quality training data for event-based stereo (Fig. 1). With our data, we then repurpose stereo foundation models to obtain a new generation of state-of-the-art, event stereo models with unparalleled generalization capabilities, which can be used in turn to further improve the original color models in challenging scenarios.

We summarize our main contributions as follows:

- We propose *EventHub*, the first framework combining neural rendering data generation and cross-modal distillation from RGB stereo foundation models to train event stereo networks without active sensor supervision.
- We demonstrate superior out-of-domain generalization compared to LiDAR-supervised models, reducing error by up to 50% on M3ED and MVSEC datasets.
- We establish bi-directional knowledge transfer between RGB and event modalities, enabling event models to improve the performance of RGB stereo foundation models in challenging nighttime conditions.

2. Related Work

Frame-based Stereo. Stereo depth estimation has transitioned from traditional hand-crafted approaches [55] to data-driven solutions [33, 52, 63]. Early learning-based methods [44, 76] focused on individual matching components, while later works [8, 30, 45, 71] introduced fully trainable pipelines combining feature extraction, cost aggregation, and disparity prediction, by leveraging the abundance of synthetic data [45, 65] for training. Building on optical flow principles, recurrent architectures [42, 66, 70] performed iterative refinement over correlation volumes. Transformer models [22, 38, 72] further enhanced matching through global attention mechanisms. Generalizing across different environments remains challenging. Solutions include learning domain-agnostic features [78, 80], incorporating geometric constraints [3, 62], self-supervised learning with photometric consistency [21, 51, 67], distillation from traditional methods [2, 10, 60] and radiance field supervision [41, 61]. Recently, foundation models trained on massive diverse datasets have demonstrated unprecedented zero-shot capabilities [6, 11, 29, 69], establishing a new state of the art. However, the scarcity of annotated stereo data in challenging conditions (e.g., night) limits their performance in such scenarios, leaving room for improvement.

Event-based Stereo. While monocular event-based depth estimation [5, 23, 34, 86] has been explored, we focus on stereo configurations exploiting binocular geometry [20]. Early stereo methods [32, 58] relied on temporal coincidence matching via frame accumulation or event-driven search, later enhanced with epipolar geometry and temporal-luminance constraints [28, 54]. Neuromorphic implementations [14, 50] deployed cooperative networks on specialized spiking hardware, while deep learning approaches [1, 64, 81] introduced learnable representations and spatio-temporal encoders. Recent works incorporate temporal context [13, 19, 24, 49], attention mechanisms [9] and unified architectures [82] that handle both stereo and optical flow. Hybrid configurations combine events with frames: binocular setups (2E+2F) [12, 47, 49] fuse modalities through recurrent networks or selection mechanisms, while asymmetric systems (1E+1F) [43, 68, 77] address cross-modal alignment challenges. Despite progress, event stereo development is severely constrained by the limited amount of annotated data [20]. Existing datasets [7, 18, 84] remain orders of magnitude smaller than frame-based ones. They also lack diversity, which constrains the ability of models to generalize beyond their training domains. This motivates us to seek alternative training strategies.

Neural Rendering for Training Data Generation. Neural radiance fields [46] enable photorealistic novel view synthesis (NVS) from sparse images, facilitating synthetic training data generation. NeRF-supervised frameworks [61] rendered stereo pairs from monocular sequences, using synthesized images and depth as proxy supervision for stereo networks. Similar approaches emerged for optical flow [41], with confidence-based filtering to remove unreliable proxy labels. Beyond depth estimation, neural rendering has been exploited for object detection [16], learning dense descriptors [74], semantic labeling [83], 6D pose estimation [35] and automated annotation in driving scenes [27]. For event cameras, video-to-event simulators [17] recycle existing datasets into synthetic streams, while specialized engines [36] generate event data with

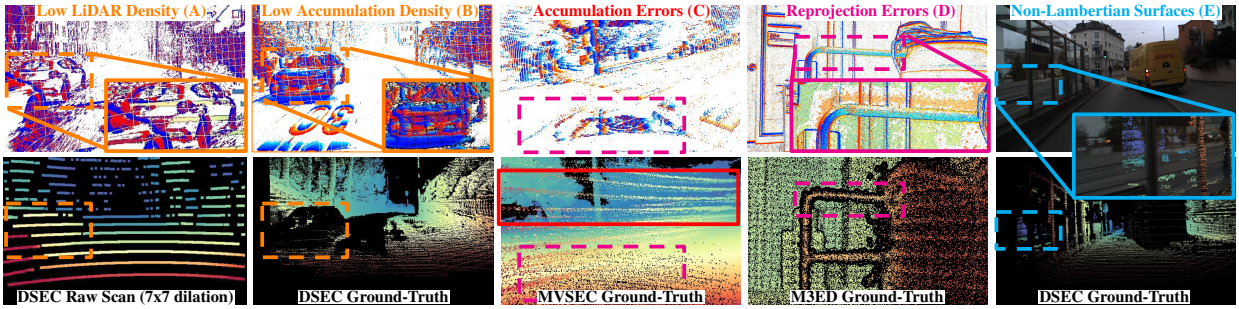


Figure 2. **Limitations of LiDAR-supervised real-world datasets.** Despite their popularity [7, 18, 84], LiDAR annotations remain sparse (A), poorly capture dynamic scenes (B–C), are prone to reprojection errors (D), and struggle on transparent or reflective surfaces (E).

optical flow labels. Concurrent work GS2E [37] generates multi-view event data from sparse RGB images via 3D Gaussian Splatting [31], and it is mildly tested for NVS and image deblurring. However, no prior work has explored neural rendering for generating event stereo training data. Leveraging efficient radiance field rendering [31, 40, 48, 59], we propose the first framework to synthesize stereo event streams from monocular RGB data, enabling large-scale training without active sensors, such as LiDAR.

3. Method

Sourcing accurate depth labels is costly and time-consuming, as it typically requires the use of active sensors such as LiDARs which, despite their high accuracy, provide very sparse data (Fig. 2 (A)). This limitation is partially mitigated by temporal accumulation; however, this strategy is ineffective in dealing with moving objects –Fig. 2 (B) or yields noise due to the motion of dynamic entities– Fig. 2 (C). Finally, imprecise calibration or non-Lambertian surfaces also harm the quality of the annotations –Fig. 2 (D,E).

Aiming to remove the dependency on noisy labeled data captured with costly LiDAR-based setups, we turn to a much cheaper data modality, simpler to obtain and already available in abundance: color images. Through the lens of RGB cameras, we can exploit state-of-the-art depth estimation techniques to annotate data with *proxy labels*, having accuracy not far from the one of LiDAR sensors. Most available color images, however, are collected by a single RGB camera, usually navigating through the scene, unpaired with any event camera counterpart. In this setting, besides proxy labels, we also need to generate *proxy events*. Conversely, when color images are paired with event data collected within the same environment, we can exploit camera calibration and multi-view geometry to annotate the real events, without the need to generate proxy events.

The overview of our framework is shown in Fig. 3. We develop techniques to extract proxy labels in the two above-mentioned settings: (i) one based on NVS frameworks,in which a modified SVRaster [59] is used to generate both proxy events and proxy labels from RGB sequences (Sec. 3.1.1), and (ii) one leveraging robust RGB stereo matching in dual RGB–Event stereo setups, where the RGB pair offers the proxy-supervision through state-of-the-art models, such as FoundationStereo [69] (Sec. 3.1.2). Moreover, to exploit the knowledge already available in the color image domain, we take a step further by exploring how to adapt pre-trained, robust RGB-based stereo matching networks [6, 69] to the event domain, thereby minimizing the need for labeled event data (Sec. 3.2).

3.1. EventHub: Data Generation

3.1.1. Synthetic Generation via Novel View Synthesis

Given sparse RGB images of a static scene, NVS frameworks [31, 46] reconstruct high-fidelity digital representations that can be rendered from arbitrary viewpoints. While NeRF-based data factories for RGB stereo exist [61], an equivalent pipeline for the event-based stereo domain remains unexplored: NVS frameworks typically output static frames rather than events, requiring fast rendering to match the event camera's temporal resolution, plus additional components to handle frames-to-events generation and motion trajectories. Therefore, we propose a novel pipeline for event stereo data generation by leveraging SVRaster [59] as the NVS foundation framework. We now describe the proposed pipeline step by step.

Image Capture and Camera Calibration. After collecting N multi-view RGB images $\hat{\mathbf{I}}_i$ of a static scene, we follow [61] and deploy COLMAP [57] to recover intrinsics $\hat{\mathbf{K}} \in \mathbb{R}^{3\times3}$ and N camera poses $[\hat{\mathbf{R}}_i|\hat{\mathbf{t}}_i] = \hat{\mathbf{T}}_i \in \mathbb{SE}(3)$.

Regularized Dense 3D Optimization. Next, for each captured scene, we fed $\hat{\mathbf{I}}_i$, $\hat{\mathbf{K}}$, and $\hat{\mathbf{T}}_i$ to SVRaster's training pipeline, obtaining a radiance representation of the scene. We follow [59] and use both MSE and SSIM to optimize the rendered image. The rendered color $\mathbf{I}$ and corresponding depth $\mathbf{Z}$ along each camera ray are defined as:

$$\mathbf{I} = \sum_{i=1}^{N} T_i \alpha_i \mathbf{c}_i, \ \mathbf{Z} = \sum_{i=1}^{N} T_i \alpha_i z_i, \ T_i = \prod_{j=1}^{i-1}(1-\alpha_j), \quad (1)$$

where $\alpha_i \in [0,1]$, $T_i \in [0,1]$, $\mathbf{c}_i \in [0,1]^3$, and $z_i > 0$ are

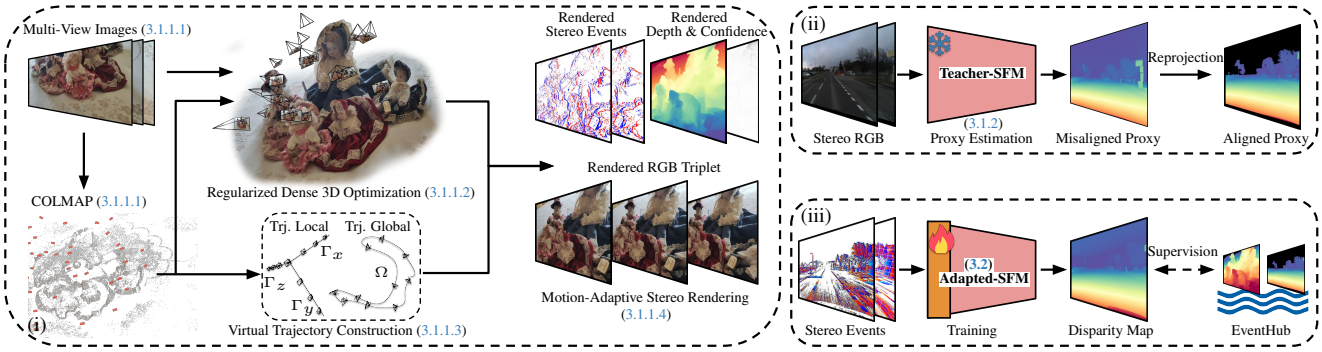


Figure 3. **Framework Overview**: We obtain training data through two complementary approaches: (i) **Event Data Factory**: SVRaster [59] generates synthetic event stereo pairs and depth labels from sparse RGB images via virtual camera trajectories (left); (ii) **Stereo Cross-Modal Distillation**: existing RGB stereo models produce proxy depth labels for real event data in calibrated RGB-Event stereo setups (top right). (iii) Both data sources are combined in EventHub to train/adapt event stereo networks (bottom right).

the opacity, the transmittance [46], the color, and the depth of the i-th voxel, respectively.

To further improve depth quality, we applied several regularizers during training: among these, (i) $\mathcal{L}_{N-\text{mean}}$ and $\mathcal{L}_{N-\text{med}}$ enforce self-consistency between rendered depth and normals, respectively aggregated using mean and median [25]; (ii) $\mathcal{L}_{\text{DAv2}}$ enforces the rendered depth to be consistent with monocular predictions from DepthAnythingV2 [73]. We studied additional regularizers $\mathcal{L}_{\text{asc}}$, $\mathcal{L}_{\text{sparse}}$, and $\mathcal{L}_{\text{mast3r}}$, with further details in the supplementary material. Each regularizer's contribution is weighted inside a regularization loss $\mathcal{L}_{\text{reg}} \doteq \lambda_{N-\text{mean}}\mathcal{L}_{N-\text{mean}} + \lambda_{N-\text{med}}\mathcal{L}_{N-\text{med}} + \lambda_{\text{asc}}\mathcal{L}_{\text{asc}} + \lambda_{\text{sparse}}\mathcal{L}_{\text{sparse}} + \lambda_{\text{DAv2}}\mathcal{L}_{\text{DAv2}} + \lambda_{\text{mast3r}}\mathcal{L}_{\text{mast3r}}$, yielding the total loss:

$$\mathcal{L} \doteq \mathcal{L}_{\text{MSE}} + \lambda_{\text{SSIM}}\mathcal{L}_{\text{SSIM}} + \mathcal{L}_{\text{reg}}. \tag{2}$$

Virtual Trajectory Construction. Given that the captured scene is static and an event camera triggers events only when the logarithmic intensity changes exceed a threshold, we emulate such variations by simulating virtual camera egomotion. We design two types of virtual trajectories: a local trajectory $\Gamma(\tau)$ and a global trajectory $\Omega(\tau)$. Both $\Gamma(\tau)$ and $\Omega(\tau)$ are continuous functions mapping a virtual time instant $\tau \in [0, 1]$ into a virtual pose $\mathbf{T}_\tau \in \mathbb{SE}(3)$. Given an initial camera pose $\hat{\mathbf{T}}_i$ (previously estimated using COLMAP), $\Gamma(\tau) = [\hat{\mathbf{R}}_i|\hat{\mathbf{t}}_i + \tau\mathbf{r}]$ applies a τ-weighted translation $\mathbf{r}$ along an arbitrary axis, *e.g.*, $\mathbf{r} = (0\ 1\ 0)^\top$. This simple setup is well-suited for object-centric captured scenes [61], where the quality of novel views tends to degrade as the rendering pose moves farther from those observed during training.

Conversely, the global trajectory $\Omega(\tau)$ is obtained by performing a least-squares fit of three cubic splines to a subset (typically one-half or one-third) of the estimated camera poses, producing smooth and continuous motion. Although a single cubic spline suffices to model the translation component $\mathbf{t}_\tau$, two additional splines followed by a re-orthogonalization ensure that $\mathbf{R}_\tau \in \mathbb{SO}(3)$. This configuration enables the synthesis of complex trajectories involving large rotations. However, to maintain meaningful viewpoints (i.e., camera orientations directed toward observed scene regions), it is generally preferable to employ this type of trajectory for indoor recordings [75].

Motion-Adaptive Stereo Rendering. After defining a virtual stereo baseline b and recovering the focal length f from the intrinsic matrix $\mathbf{K}$ of the virtual camera, we render the disparity map $\mathbf{D} = (b \cdot f)/\mathbf{Z}$ used for the supervision of stereo networks. Although depth regularization improves stability, the rendered depth maps still exhibit noise. To mitigate this, [61] proposed to extract trinocular images $\mathbf{I}_{LL}$, $\mathbf{I}_L$, $\mathbf{I}_R$ (where $\mathbf{I}_{LL}$ and $\mathbf{I}_R$ are rendered after applying respectively stereo translations $(b\ 0\ 0)^\top$ and $(-b\ 0\ 0)^\top$ to $\mathbf{t}_\tau$), balancing $\mathbf{D}$ with a trinocular photometric loss using Ambient Occlusion $\mathbf{C}_{\text{AO}}$ as the weighting term (more details in the supplementary material). To improve confidence estimation, we design a voxel-based confidence measure:

$$\mathbf{C}_{\text{Vsize}} = \text{norm}\left(\textstyle\sum_{i=1}^{N} T_i s_i\right) \odot \text{norm}\left(\textstyle\sum_{i=1}^{N} T_i \alpha_i\right), \tag{3}$$

where s_i denotes the size of the i-th voxel along the camera ray, $\odot$ represent the Hadamard product, and $\text{norm}(\cdot)$ normalizes the input to the range $[0, 1]$.

Given that SVRaster [59] does not natively support event generation, we leverage ESIM [17] to simulate stereo events from rendered frames: given two consecutive virtual frames $\mathbf{I}_L(\tau)$ and $\mathbf{I}_L(\tau + \Delta\tau)$ (or $\mathbf{I}_R(\tau)$ and $\mathbf{I}_R(\tau + \Delta\tau)$ if $(-b\ 0\ 0)^\top$ is applied), ESIM simulates the event stream along the virtual motion, assuming $\Delta\tau$ being enough small. Since both $\Gamma(\tau)$ and $\Omega(\tau)$ are continuous functions, we can render frames with arbitrary $\Delta\tau$, avoiding frame interpolation [17]; however, choosing $\Delta\tau$ is not trivial: too large values introduce simulation artifacts, while too small values leads to redundant computation. Starting from a conservative value – *e.g.*, $\Delta\tau = \frac{1}{32}$ – we dynamically adapt

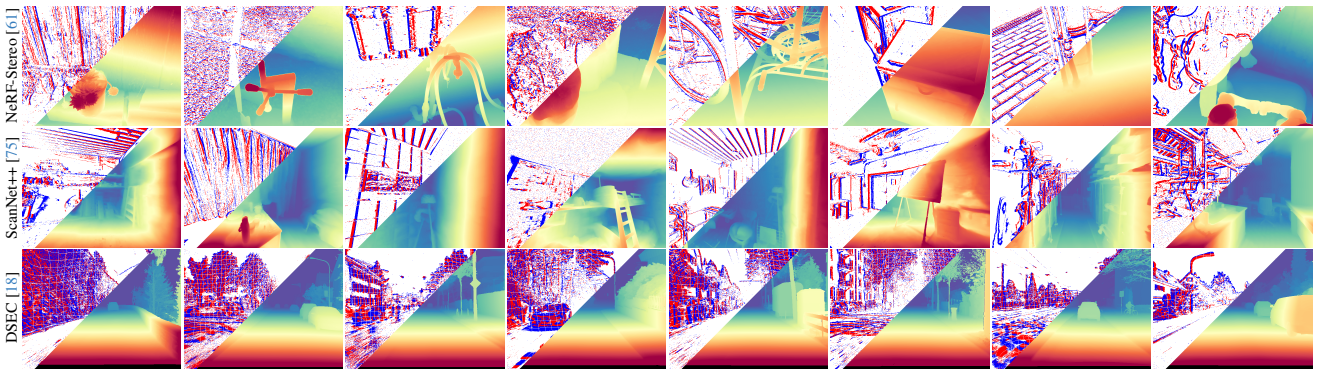


Figure 4. **Qualitative examples of events and proxy annotations by EventHub.** From top to bottom, examples obtained from NeRF-Stereo [61], ScanNet++ [75] through novel view synthesis, and from DSEC [18] through cross-modal distillation.

Dataset	MIX 1	MIX 2	MIX 3	MIX 4
NeRF-Stereo [61]	✓	✓		✓
ScanNet++ [75]		✓		✓
DSEC [18]			✓	✓

Table 1. **Combinations of datasets used by EventHub.** Proxy labels are applied to annotate each dataset.

this value using pixel motion: given the depth map $\mathbf{Z}$ and the camera poses $\mathbf{T}_\tau$ and $\mathbf{T}_{\tau+\Delta\tau}$, we compute the optical flow by projecting 3D points reconstructed from $\mathbf{Z}$ into the next view using the known relative motion $\mathbf{T}_{\tau\to\tau+\Delta\tau}$ and intrinsics $\mathbf{K}$. The flow field $\mathbf{F}$ is then obtained as the displacement between corresponding pixel projections across the two frames. To ensure bounded pixel motion and prevent event artifacts, we set the number of intermediate renderings between τ and $\tau+\Delta\tau$ to 2^n with $n = \max(\lceil \log_2(|\mathbf{F}|_{\max}) \rceil, 0)$.

3.1.2. Stereo Cross-Modal Distillation

In the second setting, we assume the availability of data collected in the same environment by two calibrated sensors: RGB and event cameras. In this case, we can obtain proxy labels from color images and transfer them to the events domain by exploiting multi-view geometry, if needed.

More specifically, as we focus on the event stereo matching task, we assume the availability of paired color-event stereo pairs $(\mathbf{I}_L, \mathbf{I}_R) - (\mathbf{E}_L, \mathbf{E}_R)$, often true for the most popular event stereo datasets available in the literature [7, 18, 84]. Accordingly, we can use an off-the-shelf, state-of-the-art Stereo Foundation Model (SFM) [6, 69] Φ_c to predict proxy labels by processing a color stereo pair $(\mathbf{I}_L, \mathbf{I}_R)$. Then, by knowing the relative pose $\mathbf{T}_{c\to e}$ between the left color camera and the left event one, we can transfer the labels and annotate the event data.

Specifically, the disparity map predicted by Φ_c is converted into depth $\mathbf{Z}_c$ by knowing baseline b_c and focal length f_c of the color stereo camera

$$\mathbf{Z}_c = (b_c \cdot f_c)/\mathbf{D}_c, \qquad \text{with} \quad \mathbf{D}_c = \Phi_c(\mathbf{I}_L, \mathbf{I}_R). \tag{4}$$

Then, being $\mathbf{u}_c$ a pixel in homogeneous coordinates on the color camera frame, we back-project it into a 3D point $\mathbf{p}_c$ according to depth $\mathbf{Z}_c(\mathbf{u}_c)$ and intrinsics $\mathbf{K}_c$. $\mathbf{p}_c$ is then expressed in the event camera reference system by applying the transformation $\mathbf{T}_{c\to e}$ between both, obtaining

$$\mathbf{p}_e = \mathbf{T}_{c\to e}\mathbf{p}_c, \qquad \text{with} \quad \mathbf{p}_c = \mathbf{Z}_c(\mathbf{u}_c)\mathbf{K}_c^{-1}\mathbf{u}_c. \tag{5}$$

Finally, we project the z coordinate of $\mathbf{p}_e$ into the event camera frame according to intrinsics $\mathbf{K}_e$. Doing this for any pixel in $\mathbf{I}_L$ yields a depth map $\mathbf{Z}_e$ aligned with $\mathbf{E}_L$. Finally, we obtain the disparity map $\mathbf{D}_e$ through triangulation, thus obtaining proxy labels $\mathbf{D}_e = (b_e \cdot f_e)/\mathbf{Z}_e$ for the event stereo pair $(\mathbf{E}_L, \mathbf{E}_R)$. In this way, we can distill the knowledge of state-of-the-art stereo models [6, 69] and reuse it in the events domain to pursue the same advances achieved in RGB stereo. This procedure is not needed if a sensor such as the DAVIS camera is available, providing pixel aligned grayscale images and event streams [84]: in such a case, the initial disparity map $\mathbf{D}_c$ already coincides with $\mathbf{D}_e$.

3.2. Repurposing RGB Stereo into Event Stereo

Besides exploiting RGB stereo models to distill proxy annotations for the event domain, we further benefit from the vast priors they learned from the abundant RGB stereo data available by repurposing the models themselves into event stereo models. In other words, we design and train an event stereo model Φ_e having the same architecture as an RGB stereo one, starting from pre-trained weights Φ_c (i.e., those used to distill proxy labels).

To this aim, keeping the number of input channels unchanged across RGB and event stereo would avoid any modification to the original deep neural network model: purposely, we encode event streams into stacked tensors according to the 3-channel Tencode representation [26] that are passed as inputs to Φ_e, by sampling events backward in time based on a fixed number of events:

$$(x, y, t, p) \to \mathbf{S}(x, y) = \begin{cases} (1,\ \frac{t_{\max}-t}{\Delta t},\ 0), & p > 0 \\ (0,\ \frac{t_{\max}-t}{\Delta t},\ 1), & p \leq 0, \end{cases} \tag{6}$$

	Training Method	SE-CFF [49]				EMatch [82]				E-FoundationStereo				E-StereoAnywhere				Avg Rank
		1PE	2PE	3PE	MAE	1PE	2PE	3PE	MAE	1PE	2PE	3PE	MAE	1PE	2PE	3PE	MAE	
(A)	Photometric [61]	88.54	71.73	55.35	7.94	92.31	69.17	44.90	3.37	93.85	73.12	49.82	3.65	92.55	72.25	51.65	4.11	6.94
(B)	EV-SceneFlow [17, 45]	66.30	50.18	41.47	3.50	71.64	53.67	42.02	3.56	61.80	48.04	41.68	3.10	64.97	49.54	41.74	3.21	6.06
(C)	MIX 1	45.61	25.15	16.54	1.87	47.17	23.23	13.58	1.70	38.70	15.63	8.83	1.39	35.86	15.13	9.03	1.36	5.00
	MIX 2	38.52	18.17	11.02	1.56	41.06	17.81	9.97	1.45	27.20	9.72	5.53	1.04	31.49	12.45	7.32	1.23	4.00
	MIX 3	24.73	8.58	5.08	1.01	31.30	11.14	5.90	1.15	20.99	6.82	4.10	0.89	24.35	8.22	4.76	0.99	2.75
	MIX 4	27.31	9.69	5.66	1.07	26.13	8.55	4.71	0.99	20.42	6.53	3.91	0.87	23.90	7.97	4.62	0.96	2.25
(D)	LiDAR (GT)	**13.82**	**4.05**	**2.37**	**0.66**	**24.11**	**7.80**	**3.99**	**0.89**	**12.53**	**3.48**	**1.98**	**0.60**	**14.66**	**4.32**	**2.51**	**0.69**	**1.00**

Table 2. **In-domain experimental results – DSEC [18] dataset.** We train four event stereo models according to different training protocols, not exploiting LiDAR annotations (A,B,C), compared against in-domain training on DSEC with LiDAR labels (D).

with $t_{\max}$ being the timestamp of the latest event occurred in the timelapse Δt during which events are stacked.

4. Experiments

4.1. Implementation and Experimental Settings

EventHub Settings. We collect proxy data from multiple sources [18, 61, 75]. For datasets without ready-to-use events [61, 75], we employ our NVS pipeline to synthesize both proxy events and depth, while for [18] we apply our distillation pipeline to estimate proxy depth only. Each NVS scene is optimized independently, setting $\lambda_{\text{SSIM}} = 0.02$, $\lambda_{N-\text{mean}} = \lambda_{N-\text{med}} = 0.0005$, $\lambda_{\text{DAv2}} = 0.01$ and disabling all other regularizers. We use three local trajectories $\Gamma_x, \Gamma_y, \Gamma_z$ (one per axis) for [61], and a global trajectory Ω for [75] with additional processing described in the supplementary material.

For NeRF-Stereo [61], we render the 270 scenes three times (one for each baseline $b \in \{0.1, 0.3, 0.5\}$) at 640×480 px resolution, setting $\Delta\tau = 0.03$. For our 403 scenes selection of ScanNet++ [75], we render both 640×480 px and 1280×720 px resolutions, each with three baselines $b \in \{0.05, 0.08, 0.1\}$, setting $\Delta\tau = 0.015$. To filter noisy labels, we adopt curation pipeline [69], training SE-CFF [49] paired with Tencode [26] on all the NVS data, discarding samples with excessive pixel errors, yielding $\sim$ 70k curated pairs. For DSEC [18] proxy labeling, we retain the train split of [4] (excluding night sequences), generating proxy labels via FoundationStereo [69] ViT-L, clipping depth between $[0.5, 100]$m before reprojection, obtaining a total of $\sim$ 30k samples. Figure 4 shows some annotated examples generated from the three datasets by EventHub, while Table 1 reports different mixtures of training data derived from them and used in our experiments.

Stereo Models and Training Settings. We evaluate two event-based stereo networks [49, 82], using Tencode [26] and VoxelGrid [85] event representations, respectively, and two RGB-based models [6, 69] adapted through our repurposing strategy and dubbed E-FoundationStereo and E-StereoAnywhere, respectively. Event networks are trained from scratch with a learning rate (lr) of $5 \cdot 10^{-4}$, while RGB models are fine-tuned from the authors' ViT-S checkpoints using lr $= 5 \cdot 10^{-5}$ and freezing the DAv2-S prior only. Training is performed in PyTorch with the AdamW optimizer, OneCycle learning rate scheduler, and data augmentations including random cropping at 576×448 px. All models are trained for 10 epochs on a single A100 GPU with batch size 2. On NVS data [61, 75] we use the NeRF-supervised loss [61], while on distilled data [18] and non-EventHub data we use the original loss of each model. These settings are used for all experiments.

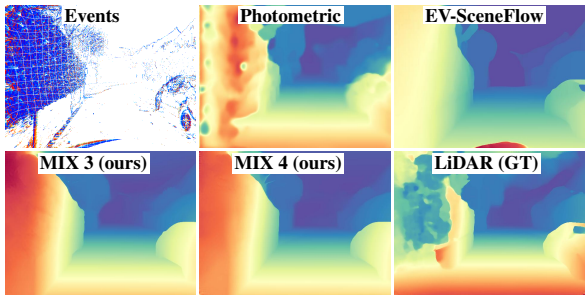


Figure 5. **Qualitative results on DSEC dataset [18].** Predictions by E-FoundationStereo trained according to different protocols.

4.2. Evaluation Datasets & Protocol

Datasets. We run our evaluation on three main datasets: DSEC [18], M3ED [7] and MVSEC [84]. DSEC features 640×480 px event stereo pairs captured by Prophesee Gen3.1 sensors, with ground truth depth annotations obtained from a 32-line LiDAR whose scans are accumulated and post-processed. We use the validation split proposed in [4] to evaluate in-domain performance. M3ED and MVSEC are instead used to evaluate the generalization performance of models trained under different paradigms (no data from these datasets is used for training). M3ED contains 1280×720 px event stereo pairs captured by Prophesee IMX636 sensors and annotated by a 64-line LiDAR. MVSEC provides 346×260 px event stereo pairs captured by DAVIS346B sensors, annotated with a 16-line LiDAR accumulated via LOAM [79].

Evaluation Metrics. We evaluate the networks using two main disparity metrics: the *Mean-Absolute-Error* (MAE) in pixels, and the percentage of pixels having an absolute disparity error larger than a specific threshold, set to 1, 2, and 3 pixels (namely, 1PE, 2PE, and 3PE).

We highlight the **best** and second-best scores.

Model	Training Method	M3ED (Day)				M3ED (Night)				M3ED (Indoor)				Avg Rank
		1PE	2PE	3PE	MAE	1PE	2PE	3PE	MAE	1PE	2PE	3PE	MAE	
SE-CFF [49]	MIX 3	46.84	25.31	16.99	2.81	58.32	36.07	24.45	3.50	51.03	31.43	23.84	4.52	2.50
	MIX 4	**35.65**	**15.18**	**9.37**	**1.22**	**51.57**	**26.84**	**15.33**	**1.70**	48.56	**27.36**	**19.33**	**2.95**	**1.08**
	LiDAR (GT)	58.82	41.41	32.93	3.05	58.94	36.78	23.55	2.20	**45.33**	28.39	21.59	4.48	2.42
EMatch [82]	MIX 3	86.18	77.61	72.13	40.36	90.02	84.94	82.26	45.41	76.99	65.23	58.98	15.80	3.00
	MIX 4	**43.99**	**20.87**	**12.93**	**2.23**	**63.69**	**38.74**	**26.38**	**5.03**	**58.42**	**32.16**	**21.48**	**3.10**	**1.00**
	LiDAR (GT)	83.16	71.65	62.81	12.22	83.36	73.03	66.06	18.63	64.34	46.85	38.80	7.95	2.00
E-FoundationStereo	MIX 3	33.44	19.20	12.37	1.49	49.26	26.09	14.94	1.84	40.27	22.08	15.73	**2.37**	2.00
	MIX 4	**26.38**	**11.57**	**6.96**	**0.98**	**46.90**	**23.09**	**12.96**	**1.54**	40.74	**21.83**	**15.61**	2.45	**1.25**
	LiDAR (GT)	54.80	39.48	31.43	2.89	55.77	34.93	22.60	1.99	**38.93**	22.03	15.96	2.87	2.75
E-StereoAnywhere	MIX 3	47.53	27.85	18.83	4.48	59.21	34.29	21.21	3.64	46.02	26.71	19.55	3.19	2.42
	MIX 4	**34.99**	**13.01**	**7.88**	**1.12**	**58.62**	**27.85**	**14.42**	**1.71**	**42.74**	**23.79**	**16.84**	**2.58**	**1.00**
	LiDAR (GT)	63.70	43.33	33.90	3.26	63.23	41.22	26.72	2.78	44.27	25.81	18.80	3.72	2.58

Table 3. **Out-of-domain experimental results – M3ED [7] dataset.** We compare the generalization capability of the four event stereo models trained with MIX 3 and MIX 4 against their counterparts trained using DSEC LiDAR labels.

Model	Training Method	MVSEC (Day)				MVSEC (Night)				MVSEC (Indoor)				Avg Rank
		1PE	2PE	3PE	MAE	1PE	2PE	3PE	MAE	1PE	2PE	3PE	MAE	
SE-CFF [49]	MIX 3	77.13	53.87	31.12	3.21	78.19	54.75	31.24	3.64	42.14	24.53	18.04	3.30	2.17
	MIX 4	**31.99**	**12.00**	**6.88**	**1.11**	**40.54**	**19.00**	**10.13**	**1.45**	**29.66**	**12.34**	**6.89**	**1.39**	**1.00**
	LiDAR (GT)	97.82	94.85	90.47	6.12	96.89	92.95	87.14	6.07	46.67	26.37	15.43	1.78	2.83
EMatch [82]	MIX 3	93.80	80.23	59.05	6.00	91.74	75.37	49.51	4.67	58.44	35.05	24.24	3.02	2.08
	MIX 4	**56.29**	**21.61**	**6.67**	**1.39**	**68.51**	**40.48**	**14.92**	**1.81**	**46.03**	**21.40**	**12.36**	**1.93**	**1.00**
	LiDAR (GT)	99.47	98.36	95.83	6.70	98.56	96.20	92.40	6.21	66.21	43.21	28.13	2.60	2.92
E-FoundationStereo	MIX 3	81.78	54.75	28.95	2.75	81.89	58.54	36.27	2.73	34.45	18.51	12.48	1.62	2.08
	MIX 4	**45.94**	**20.92**	**9.45**	**1.33**	**58.15**	**38.14**	**18.02**	**1.75**	**24.55**	**9.11**	**5.29**	**1.07**	**1.00**
	LiDAR (GT)	97.91	94.65	89.45	6.04	97.32	94.15	89.74	6.26	40.19	21.27	12.62	1.61	2.92
E-StereoAnywhere	MIX 3	77.22	60.04	42.41	4.37	75.97	57.26	35.88	3.47	40.68	24.74	19.64	4.18	2.08
	MIX 4	**68.18**	**44.60**	**20.92**	**1.96**	**72.21**	**47.93**	**20.39**	**1.96**	**21.27**	**7.84**	**4.48**	**0.94**	**1.00**
	LiDAR (GT)	98.39	95.68	90.43	7.85	97.59	94.77	89.27	8.12	49.80	31.71	21.39	2.79	2.92

Table 4. **Out-of-domain experimental results – MVSEC [84] dataset.** We compare the generalization capability of the four event stereo models trained with MIX 3 and MIX 4 against their counterparts trained using DSEC LiDAR labels.

4.3. In-Domain Evaluation

We first assess how the different mixtures of data generated by EventHub impact the accuracy of trained models, as well as comparing our training strategies with existing LiDAR-free alternatives as well as with LiDAR supervision. Table 2 collects the outcome of this evaluation, carried out by training four event stereo models according to four main strategies. The first two rows report results obtained by training the models: (A) using photometric loss between DSEC's RGB stereo images projected into event frame, or (B) using a synthetic event dataset derived from SceneFlow [45] via [17], which provides perfect ground truth disparities, yet proxy event data. Both approaches are scarcely effective, with MAE never dropping below 3 pixels for any model.

Then, we report the results achieved by training on the data and annotation produced by EventHub (C), involving different mixtures of data. Notably, MIX 1 already yields largely lower error values, benefiting from the stronger supervision of the proxy labels rendered by SVRaster. Adding data from ScanNet (MIX 2) yields moderate improvements. Training on MIX 3 further boosts performance across all stereo models, which is not surprising since it involves training data from the same domain used in the evaluation. Nonetheless, combining all data sources (MIX 4) produces the best overall performance for EMatch [82], E-FoundationStereo and E-StereoAwywhere. The bottom row reports the accuracy obtained by running in-domain supervised training using LiDAR ground truth (D), which unsurprisingly yields the lowest errors, yet proving how models trained with MIX 4 get very close to this upper bound despite not using any ground truth annotation from LiDAR.

Despite the thoroughness of Tab. 2, the shortcomings of LiDAR data for both training and evaluation are fully not apparent. Figure 5 unveils how the sparse nature of LiDAR annotations hampers the network's ability to produce dense and accurate disparity maps. In contrast, training with MIX 3 already avoids most of the artifacts introduced by LiDAR supervision, even though this is not reflected in the error metrics, which are based on LiDAR data.

4.4. Out-of-Domain Evaluation

We now extend our evaluation beyond the single domain represented by DSEC, moving to M3ED and MVSEC. These cover both indoor and outdoor scenarios and are collected by sensors with very different properties, thus representing a significant domain shift with respect to DSEC.

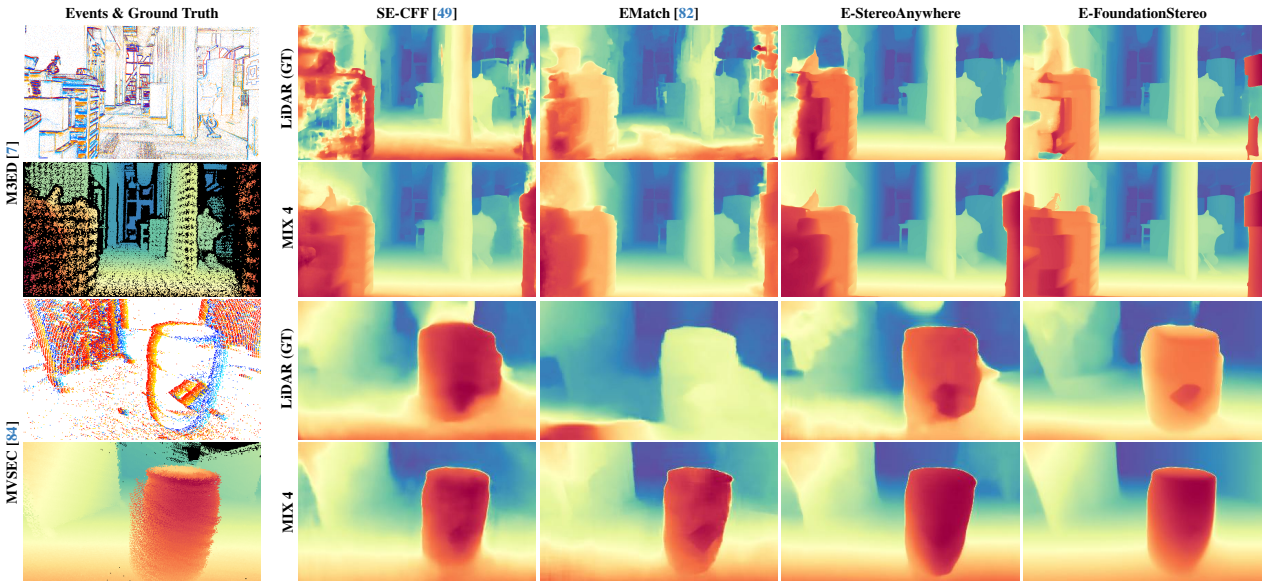


Figure 6. **Qualitative results on M3ED [7] and MVSEC [84].** Predictions by the four models trained with LiDAR labels or MIX 4.

Model	Training Method	DSEC (Night) 1PE	2PE	3PE	MAE
FoundationStereo (ViT-S)	Author's Weights [69]	68.40	47.69	35.80	3.89
	MIX 3 (Night)	**24.75**	**8.09**	**4.25**	**1.01**
FoundationStereo (ViT-L)	Author's Weights [69]	30.06	14.87	10.88	1.87
	MIX 3 (Night)	**25.33**	**8.48**	**4.56**	**1.06**
StereoAnywhere (ViT-S)	Author's Weights [6]	33.01	14.24	8.93	1.61
	MIX 3 (Night)	**30.61**	**10.81**	**5.72**	**1.22**
StereoAnywhere (ViT-L)	Author's Weights [6]	**31.34**	13.20	8.27	1.52
	MIX 3 (Night)	32.42	**11.45**	**5.93**	**1.23**

Table 5. **Experimental results on DSEC night images [18] – RGB stereo models.** Fine-tuning SFMs on proxy labels derived from our event models yields improvements on nighttime images.

Table 3 presents the results achieved by the four models on M3ED, each supervised with MIX 3, MIX 4 and LiDAR training strategies, i.e., the same models trained on DSEC and transferred to M3ED without any additional fine-tuning. Since MIX 1 and MIX 2 perform worse than MIX 3 and MIX 4, they are omitted from here onward. We report results averaged over three main subdomains: `Day`, `Night` and `Indoor` scenes. We observe that models trained with MIX 4 largely outperform their counterparts trained with LiDAR annotations in terms of generalization. This confirms the sub-optimality of supervision provided by sparse and noisy LiDAR measurements, which is often surpassed even by MIX 3 alone (which just replaces LiDAR annotations with proxy labels, without additional training data).

Table 4 reports the outcome of the evaluation on MVSEC, using the same four models trained with MIX 3, MIX 4, or LiDAR labels, averaging results over the same three subdomains as before. Once again, MIX 4 emerges as the absolute winner in terms of generalization, while MIX 3 consistently achieves the second-best results, except in a few cases. Finally, Fig. 6 shows examples from M3ED and MVSEC datasets, highlighting that any model produces much sharper and more accurate disparity maps when trained with MIX 4 than with LiDAR labels.

4.5. Closing the Loop: Improving SFMs at Night

Finally, we investigate whether the event-based stereo models trained with proxy labels can, in turn, serve as sources of new proxy labels for annotating color images in scenarios where conventional SFMs struggle, such as nighttime conditions. To this end, we generate proxy labels from stereo pairs $(\mathbf{E}_L, \mathbf{E}_R)$ and transfer them to $(\mathbf{I}_L, \mathbf{I}_R)$, reversing the procedure described in Sec. 3.1.2.

Table 5 presents the results of this experiment, showing that both FoundationStereo and StereoAnywhere perform poorly on nighttime images. After fine-tuning them for 10 epochs on the proxy labels predicted by their E-FoundationStereo and E-StereoAnywhere couterparts, their accuracy improves substantially, effectively closing the loop across modalities.

5. Conclusion

We presented *EventHub*, a paradigm for supervising deep event-stereo networks that does not rely on expensive, yet noisy, annotations from LiDAR sensors. *EventHub* leverages novel view synthesis and knowledge distillation to obtain proxy labels (and proxy events, when needed) directly from conventional color image collections. Models trained with *EventHub* achieve superior generalization, outperforming those trained with LiDAR labels in cross-domain scenarios. These models can be used to close the loop across modalities, yielding proxy labels to improve RGB stereo models in scenarios where they struggle.

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